

**Industrial Internship Report on
"HumanResourceManagementSystem"**

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was Human Resource Management System

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

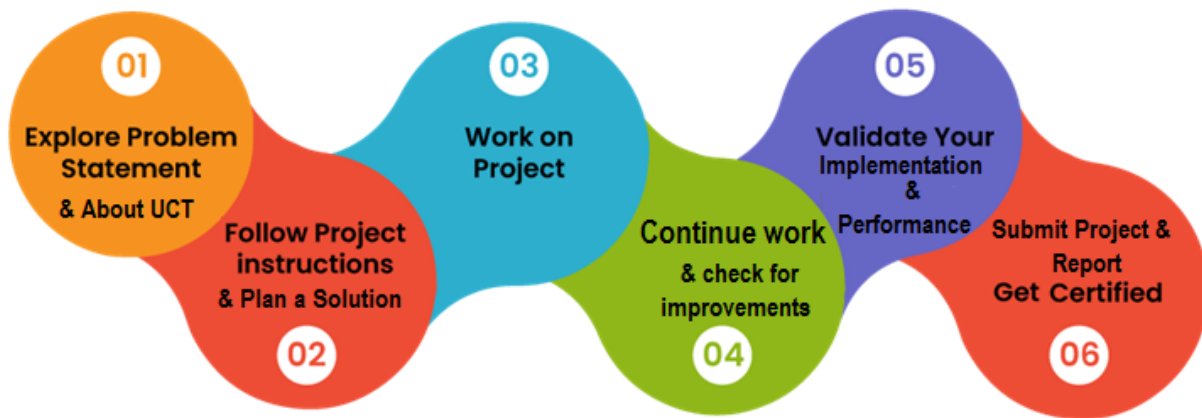
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all my brother Arpit Lekhwar , teacher Sir Aashish and my mother Pooja Lekhwar,

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LRaWAN), Java Full Stack, Python, Front end** etc.



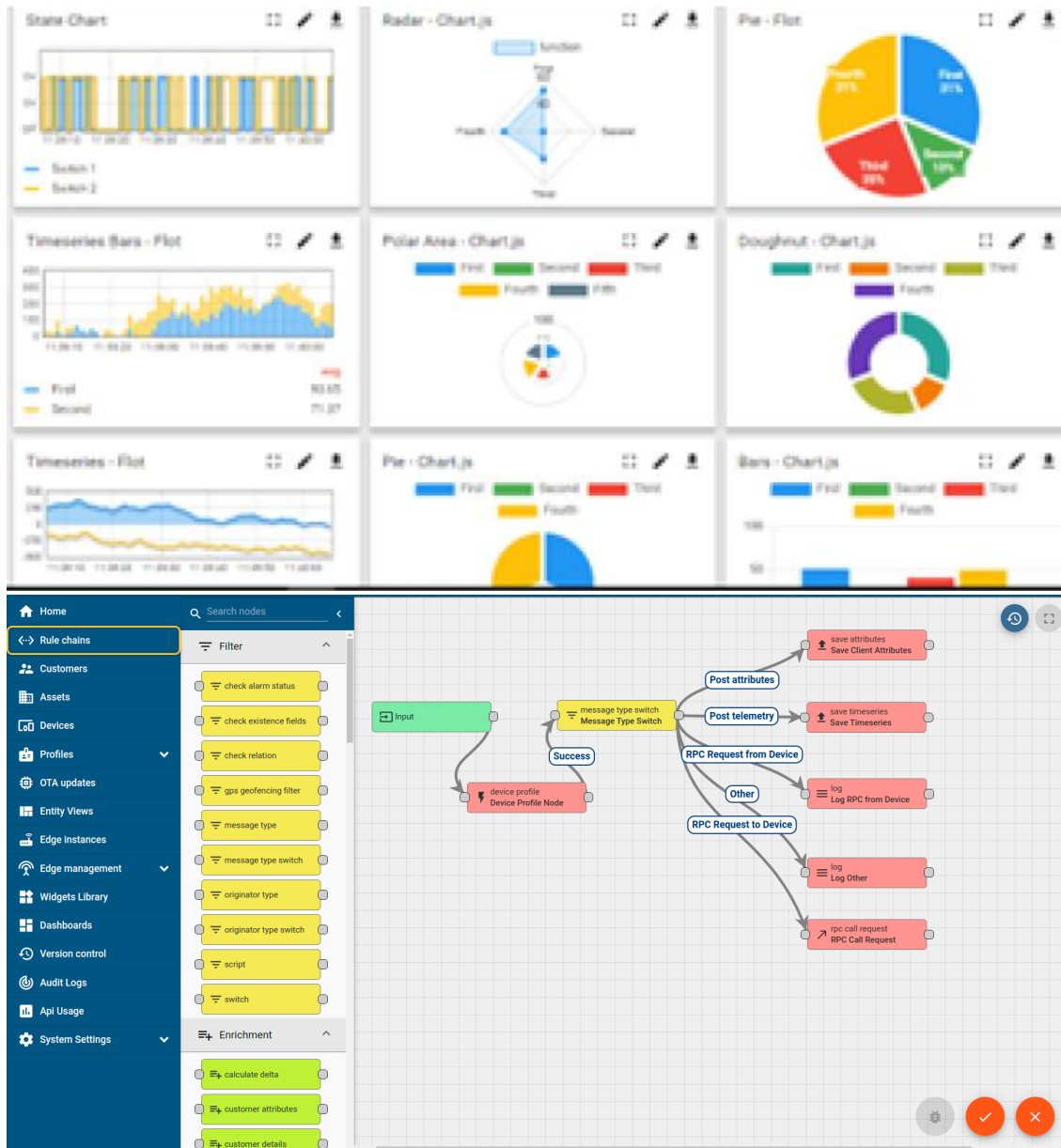
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY **WATCH**

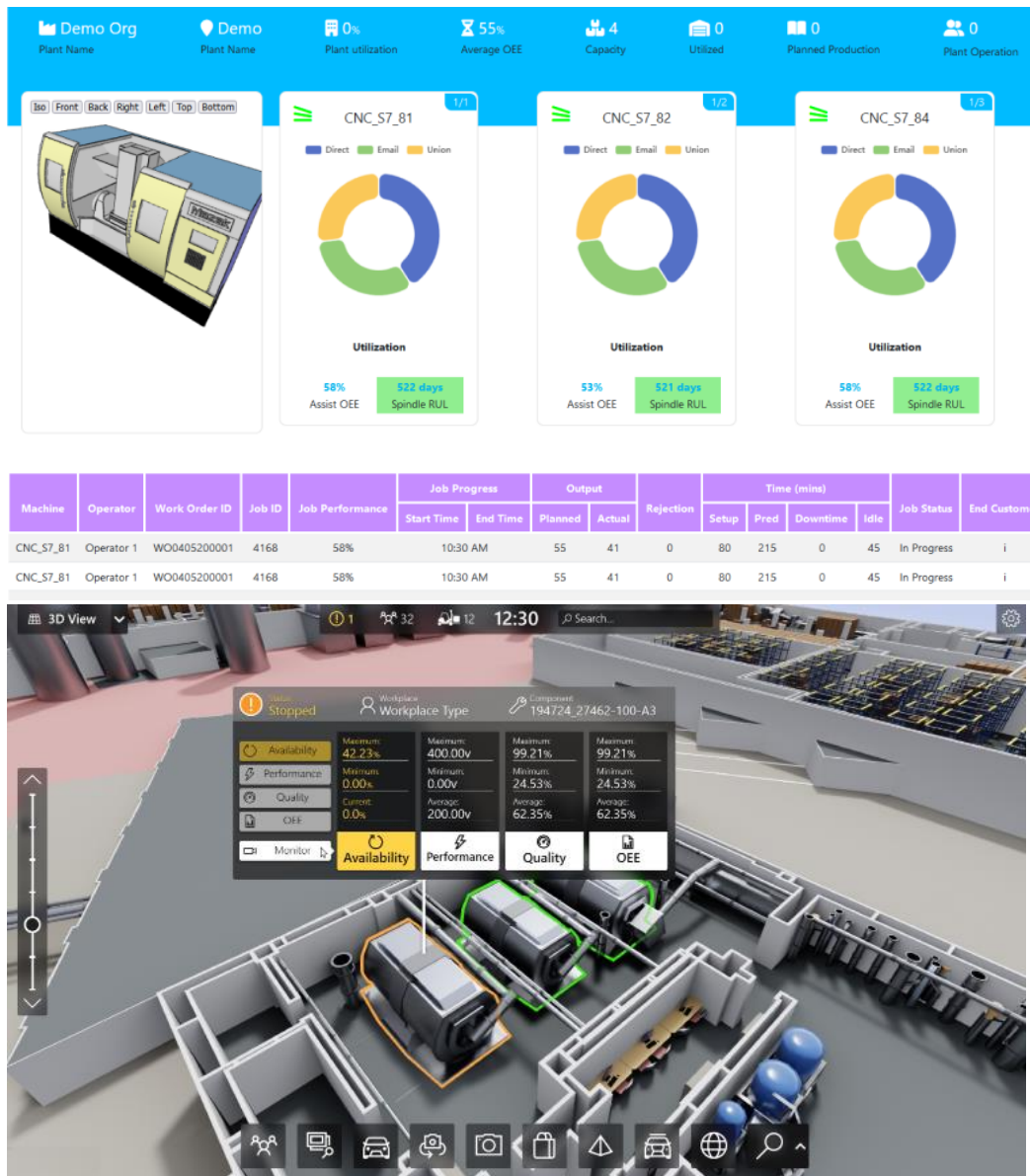
ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



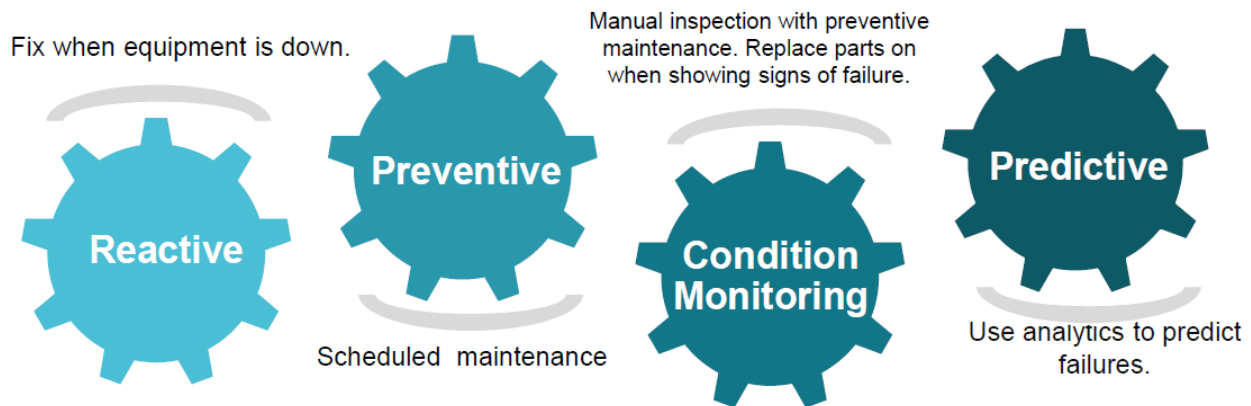


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

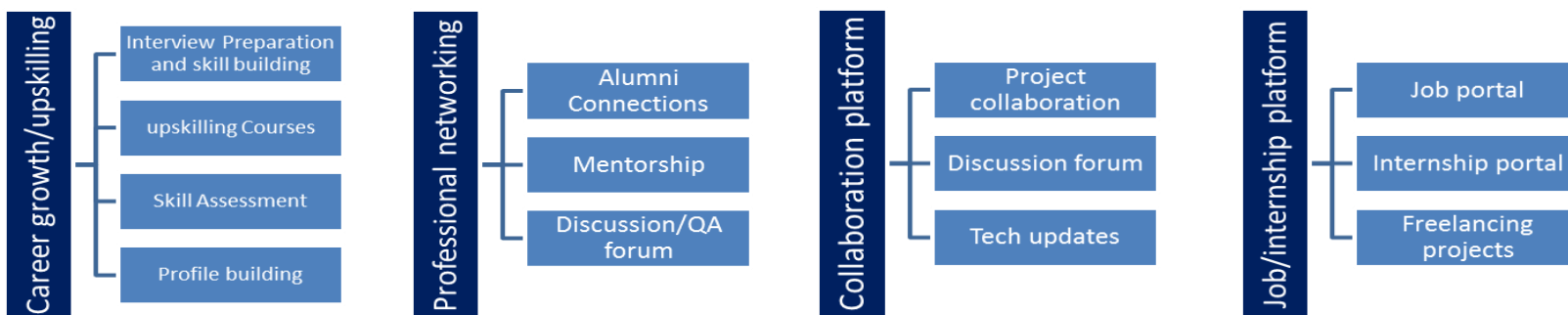


2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.





2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

3 Problem Statement

In the assigned problem statement

My problem statement is HRMS human resource management system.

In current time we are in grave need to develop such systems that help people with their daily day to day work. That is where this system comes into use.

The human resource management system that I have designed is basically one stop solution to the current management of human systems. It includes employee data management like employee id, designation, department, and contact information.

It also comprises of attendance tracking feature allowing hassle free attendance without much putting in much efforts.

Since it comprises of all the data arranged in definite structure it helps efficient searching of data of all the employees.

4 Existing and Proposed solution

1. Spreadsheet & Manual Systems (e.g., Excel, Google Sheets, Physical Registers)

- **Summary:** Small organizations often rely on manual spreadsheets to record employee lists, daily attendance, and leave tracking.
- **Limitations:**
 - High risk of data entry errors, accidental deletion, and data redundancy.
 - Lack of built-in role authentication or access restrictions.
 - Difficult to run cross-referencing queries (e.g., verifying if an employee exists before approving a leave).

2. Commercial Enterprise HRMS / ERP Software (e.g., SAP SuccessFactors, Workday, Zoho People)

- **Summary:** Cloud-based, fully integrated human resource management suites designed for enterprise environments.
- **Limitations:**
 - Prohibitive licensing and deployment costs for small teams or academic prototyping.
 - Heavy resource overhead, steep learning curves, and complex server/database dependencies.
 - Over-engineered for scenarios requiring rapid, standalone local deployment.

3. Simple In-Memory Academic Prototypes

- **Summary:** Basic Java CLI programs created using standard collections (ArrayList or HashMap) to demonstrate object-oriented programming concepts.
- **Limitations:**
 - **Session Volatility:** Data is lost immediately upon program termination due to lack of persistence.
 - **Lack of Input Sanitization:** Frequent program crashes due to unhandled InputMismatchException or invalid date formats.
 - **Tight Coupling:** Business logic, input handling, and data models are often mixed inside a single method, reducing maintainability.
- **Proposed Solution: Core Java HRMS Prototype**

The proposed solution is a **Console-Based Human Resource Management System (HRMS)** built in modular Core Java that balances lightweight execution with full lifecycle data management:

- **Authentication Layer:** Secured admin login prompt restricting unauthorized access to management operations.
- **Employee Management Module:** Full CRUD (Create, Read, Update, Delete) capability with unique employee ID enforcement.
- **Attendance Tracking Module:** Date-wise and employee-wise logging for tracking "Present", "Absent", and "On Leave" statuses.
- **Leave Management Lifecycle:** Structured leave application and administrative review pipeline (Approve/Reject actions).
- **Search & Analytics Engine:** Multi-attribute filtering (search by ID, Name, or Department) and executive summary reports displaying workforce metrics.
- **File-Based Persistence:** Automatic data persistence using Java Object Serialization (hrms_data.ser) to retain system state across application restarts without requiring external database server installation.

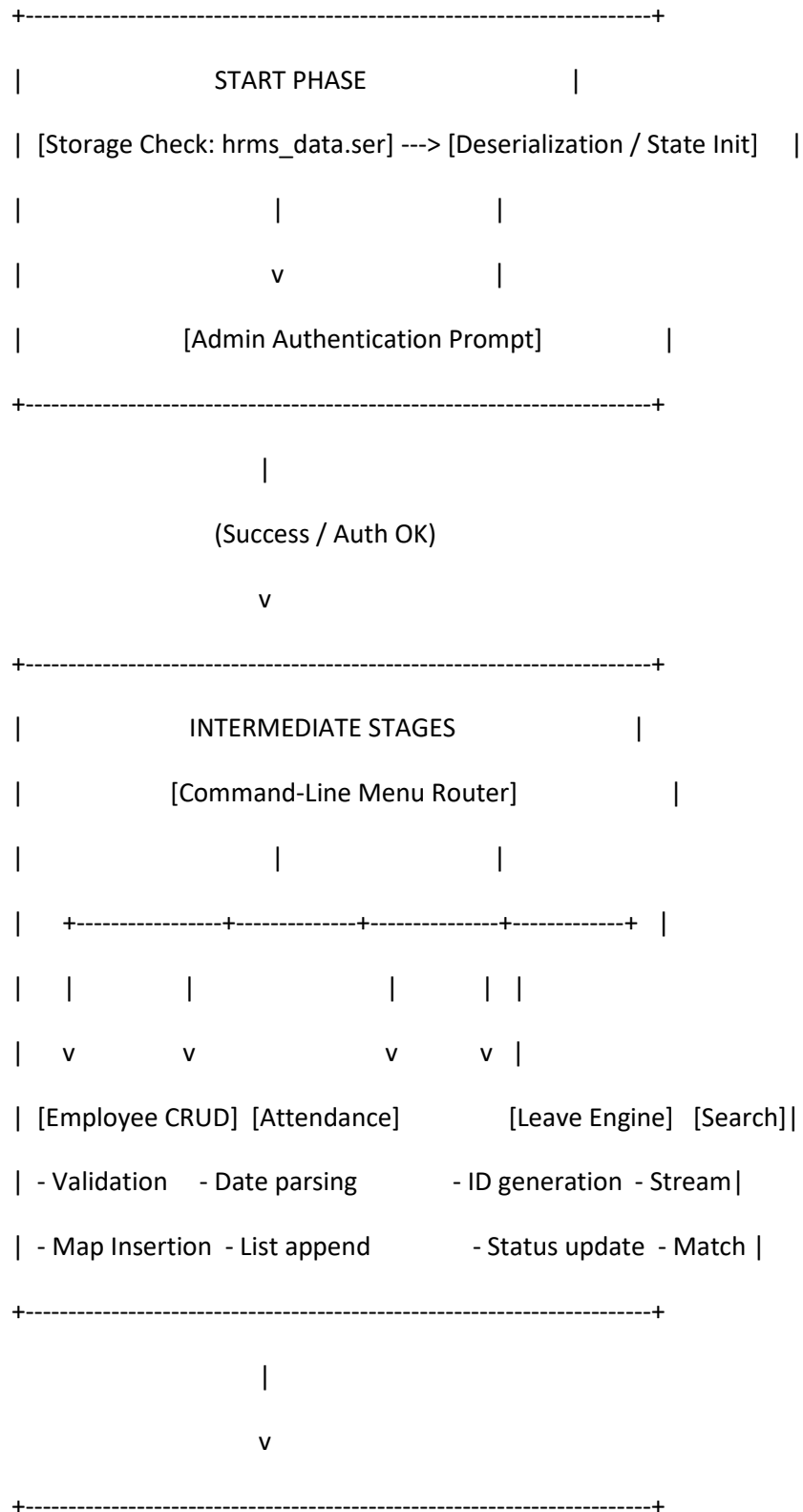
4.1 Code submission (Github link)

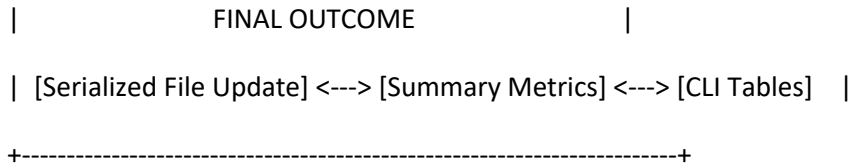
<https://github.com/KshitizLekhwar/upskillcampus/blob/main/HumanResourceManagementSystem.java>

4.2 Report submission (Github link) :

https://github.com/KshitizLekhwar/upskillcampus/blob/main/HumanResourceManagementSystem_Kshitiz_USC_UCT.pdf

5 Proposed Design/ Model





6 Phase 1: Start (System Initialization & Verification)

1. Storage Detection & State Loading (Deserialization)

- The program checks the working directory for the persistent binary file `hrms_data.ser`.
- **Branch A (Existing File):** Reads the serialized container `HRMSData` via `ObjectInputStream` and hydrates the in-memory collections (`employees`, `attendanceList`, `leaveRequests`).
- **Branch B (First Run):** Instantiates an empty `HRMSData` object with initialized hash maps and lists.

2. Access Control & Authentication

- Prompts the administrator for credentials (`admin / admin123`).
- Employs an attempt-limiter loop (maximum 3 attempts). A successful match grants access to the main loop; failure terminates execution immediately to prevent unauthorized state access.

7 Phase 2: Intermediate Stages (Processing, Routing & State Mutation)

Once authenticated, the system enters an interactive event-driven console loop that routes control based on user selection:

Input Stream ---> Input Sanitization ---> Business Rule Validation ---> Collection Mutation

8 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

9 My learnings

- **Core OOP & Modular Design:** Applied encapsulation across distinct entities (Employee, AttendanceRecord, LeaveRequest) and separated business logic from data storage using Java Collections (HashMap, ArrayList).
- **Data Persistence:** Implemented Java Object Serialization (ObjectOutputStream/ObjectInputStream) to save and restore application state via hrms_data.ser without external database dependencies.
- **Defensive Programming:** Handled run-time exceptions (NumberFormatException, DateTimeParseException) to prevent crashes and validate date formats using java.time.LocalDate.
- **Business Logic & State Management:** Built duplicate-key validation, leave-request status workflows (PENDING → APPROVED/REJECTED), and multi-field search filtering.
- **CLI Data Presentation:** Formatted console output into clean, aligned ASCII tables using printf / String.format and computed summary metrics using Java Streams.

10 Future work scope

I was not able to develop this as interface in gui as I did not know how to use java swing , spring and springboot , due to which I was completely unaware of how to develop gui applications.